CHAPTER 1

Interpolating with Polynomials

Let's say someone on a distant planet records video of a hammer being thrown up into the air. They send you three random frames of the hammer in flight. Each frame has a timestamp, and you can clearly see how high the hammer is in each one. Can you create a 2nd degree polynomial that explains the entire flight of the hammer?

In other words, you have three points $(t_0, h_0), (t_1, h_1), (t_2, h_2)$. Can you find a, b, c such that the graph of $at^2 + bt + c = t$ passes through all three points?

The answer is yes. In fact, given any n points[1], there is exactly one $n-1$ degree polynomial that passes through all the points.

There are a lot of variables floating around. Let's make it concrete: The photos are taken at $t = 2$ seconds, $t = 3$ seconds, and $t = 4$ seconds. In those photos, the height of the hammer is 5m, 7m, and 6m. So, we want our polynomial to pass through these points: (2, 5), (3, 7), (4,6).

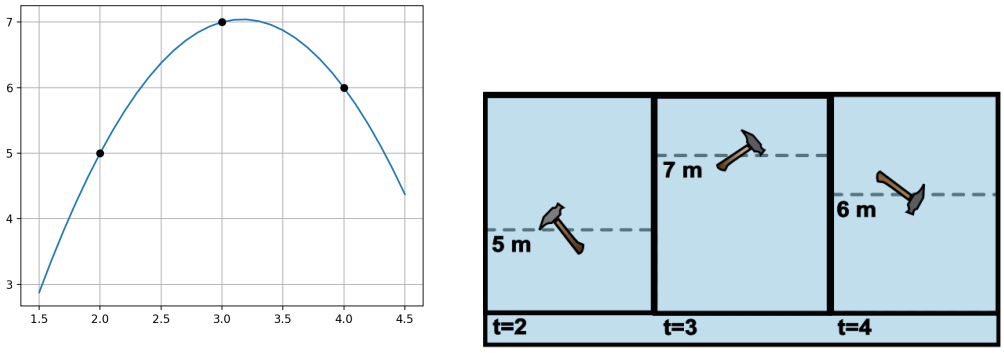


Figure 1.1: Left: the graph of the hammer going through the given points. Right: the three hammer frames.

How can you find that polynomial? Let's do it in small steps. Can you create a 2nd degree polynomial that is not zero at $t = 2$, but is zero at $t = 3$ and $t = 4$? Yes, you can: $(x-3)(x-4)$ has exactly two roots at $t = 3$ and $t = 4$. The value of this polynomial at $t = 2$ is $(2-3)(2-4) = 2$. We really want it to be 5m, so we can divide the whole polynomial by 2 and multiply it by 5.

[1] $n = 2$ will result in a line in slope intercept point, and $n = 1$ is a single point

Now we have the polynomial:

$$f_0(x) = \frac{5}{(2-3)(2-4)}(x-3)(x-4) = \frac{5}{2}x^2 - \frac{35}{2}x + 30$$

This is a second degree polynomial that is 5 at $t = 2$ and 0 at $t = 3$ and $t = 4$.

Now, we create a polynomial that is 7 at $t = 3$ and 0 at $t = 2$ and $t = 4$:

$$f_1(x) = \frac{7}{(3-2)(3-4)}(x-2)(x-4) = -7x^2 + 42x - 56$$

Finally, we create a polynomial that is 6 at $t = 4$ and zero at $t = 2$ and $t = 3$:

$$f_2(x) = \frac{6}{(4-2)(4-3)}(x-2)(x-3) = 3x^2 - 15x + 18$$

Adding these three polynomials together gives you a new polynomial that touches all three points:

$$f(x) = \frac{5}{2}x^2 - \frac{35}{2}x + 30 - 7x^2 + 42x - 56 + 3x^2 - 15x + 18 = -\frac{3}{2}x^2 + \frac{19}{2}x - 8$$

You can test this in Python with your `Polynomial` class. Create a file called `test_interpolation.py`. Add this code:

```
from Polynomial import Polynomial
import matplotlib.pyplot as plt

in_x = [2,3,4]
in_y = [5,7,6]

pn = Polynomial([-8, 19/2, -3/2])
print(pn)

# These lists will hold our x and y values
x_list = []
y_list = []

# Starting x
current_x = 1.5

while current_x <= 4.5:
    # Evaluate pn at current_x
    current_y = pn(current_x)

    # Add x and y to respective lists
    x_list.append(current_x)
    y_list.append(current_y)
```

```
24
25      # Move x forward
26      current_x += 0.05
27
28  # Plot the curve
29  plt.plot(x_list, y_list)
30
31  # Plot black circles on the given points
32  plt.plot(in_x, in_y, "ko")
33  plt.grid(True)
34  plt.show()
```

You should get a nice plot that shows the graph of the polynomial passing through those three points.

In general, then, if you provide any three points $(t_0, h_0), (t_1, h_1), (t_2, h_2)$, there is a second degree polynomial that pass through all three:

$$\frac{h_0}{(t_0-t_1)(t_0-t_2)}(x-t_1)(x-t_2)+\frac{h_1}{(t_1-t_0)(t_1-t_2)}(x-t_0)(x-t_2)+\frac{h_2}{(t_2-t_0)(t_2-t_1)}(x-t_0)(x-t_1)$$

What if you are given 9 points $((t_0, h_0), (t_1, h_1), \ldots, (t_8, h_8))$ and want to find a 8th degree polynomial that passes through all of them? Just what you would expect:

$$\frac{h_0}{(t_0-t_1)(t_0-t_2)\ldots(t_0-t_8)}(x-t_1)(x-t_2)\ldots(x-t_8)+\ldots+\frac{h_8}{(t_8-t_0)\ldots(t_8-t_7)}(x-t_0)\ldots(x-t_7)$$

FIXME: Do I need to define summation and prod here?

The general solution is, given n points, the $n-1$ degree polynomial that goes through them is

$$y = \sum_{i=0}^{n}\left(\prod_{\substack{0\le j\le n\\ j\neq i}}\frac{x-t_j}{t_i-t_j}\right)h_i$$

That would be tedious for a person to compute, but computers are perfect for this stuff. Let's create a method that creates instances of Polynomial using interpolation.[2]

1.1 Interpolating polynomials in python

Your method will take two lists of numbers: one contains x-values and the other contains y-values. So comment out the line that creates the polynomial in `test_interpolation.py` and create it from two lists:

[2]Note that other textbooks or resources may refer to this as *Lagrange Polynomial Interpolation*.

```
in_x = [2,3,4]
in_y = [5,7,6]
# pn = Polynomial([-8, 19/2, -3/2])
pn = Polynomial.from_points(in_x, in_y)
print(pn)
```

Add the following method to your Polynomial class in `Polynomial.py`

```
@classmethod
    def from_points(cls, x_values, y_values):
        coef_count = len(x_values)

        # Sums start with a zero polynomial
        sum_pn = Polynomial([0.0] * coef_count)
        for i in range(coef_count):

            # Products start with the constant 1 polynomial
            product_pn = Polynomial([1.0])
            for j in range(coef_count):

                # Must skip j=i
                if j != i:
                    # (1x - x_values[j]) has a root at x_values[j]
                    factor_pn = Polynomial([-1 * x_values[j], 1])
                    product_pn = product_pn * factor_pn

            # Scale so product_pn(x_values[i]) = y_values[i]
            scale_factor  = y_values[i] / product_pn(x_values[i])
            scaled_pn = scale_factor * product_pn

            # Add it to the sum
            sum_pn = sum_pn + scaled_pn

        return sum_pn
```

It should work exactly the same as before. You should get the same polynomial printed out as before as well as the same plot of the curve passing through the three points.

How about five points? Change `in_x` and `in_y` at the start of `test_interpolation.py`:

```
in_x = [1.7, 2, 2.7, 3.5, 4, 4.4]
in_y = [8, 12, 1, 4, -1, 6]
```

You should get a polynomial that passes through all five points:

$$11.21x^5 - 171.05x^4 + 1019.44x^3 - 2957.53x^2 + 4161.78x - 2258.75$$

It should look like this:

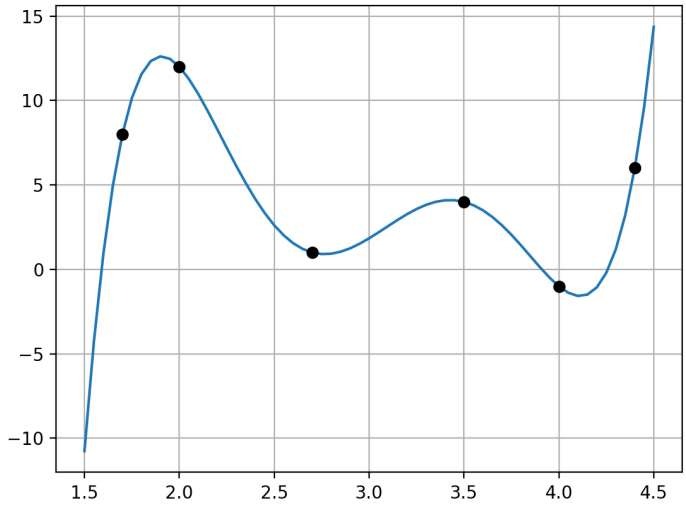


Figure 1.2: 5 point interpolation of points.

This is a draft chapter from the Kontinua Project. Please see our website (https://kontinua.org/) for more details.

Appendix A

Answers to Exercises